Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/26/2006
PI Name:	Snell, Edward H. Ph.D.		
Project Title:	Cool Crystals–A Physical and Biochemical Study of Macromolecular Crystal Cyropreservation		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14203-1102	**Congressional District:**	26
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	09/01/2001	**End Date:**	11/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Judge, R. (Abbott Laboratories) van der Woerd, M. (BAE Systems)		
Grant/Contract No.:	NAG8-1928		
Performance Goal No.:			
Performance Goal Text:			

Cryocooling is accomplished by appropriate choice of cryoprotectant and correct crystal handling technique. For characterization of the interactions between the cryoprotectant and the macromolecules, we will use techniques including dynamic light scattering and circular dichroism. These techniques will allow us to look at macromolecular aggregation through dispersion in solution and at the effects of the cryoprotectant on the secondary structure of the sample. X-ray methods, including structural studies, reciprocal space mapping and topography, will be used to examine the effect of cryoprotectant on the crystals. This will be carried out both at room temperature using flow cell techniques and under cryogenic conditions. Different cryopreservation techniques will also be examined. Cooling and the effects of the X-ray beam on the cooled crystal will be studied. Mass spectrometry will be used on the irradiated samples to assess how well free radical migration has been suppressed. A physical model of cryopreservation will be built up from the X-ray and biochemical results. This model will be tested and used to further optimize the cryogenic techniques. As part

Task Description:	of the microgravity application of cryocooling envisaged for the International Space Station we will also look at long term storage of cooled samples. The optimized cryogenic conditions will be used with microgravity grown samples to obtain the best possible structural data on a variety of samples of biochemical interest and interest to NASA and the scientific community at large. Rapid cooling (or cryopreservation) of macromolecular crystals is routinely used for structural X-ray data collection. Cryopreservation decreases the thermal scatter and radiation damage resulting in increased resolution and completeness of the data. Microgravity dramatically reduces the mosaicity of macromolecular crystals, while cryocooling increases this property. We aim to cryocool microgravity and ground grown crystals in such a way as to maintain their physical perfection and optimize the structural data. Specifically we aim to: 1. Study biochemical effects of cryoproctectants on the samples. 2. Use laboratory X-ray methods to examine effects on crystals at room temperature. 3. Use laboratory X-ray and infra-red methods to examine choice of cryoprotectant and technique under cryoconditions. 4. Characterize the radiation decay biochemically. 5. Produce a model of the physical cryopreservation process. 6. Study storage at different cryogenic temperatures. 7. Use the optimized techniques for both microgravity and structural data collection. A detailed systematic study of cryogenic techniques for macromolecular crystal preservation including an assessment of radiation damage will benefit the macromolecular crystal growth community in many ways. For long-term missions on the ISS, macromolecular crystal samples will need to be protected against degradation until they can be analyzed/used. In addition, to make full use of the X-ray facilities on earth, i.e. third generation synchrotron sources, the samples will have to be cryogenically preserved in the X-ray beam. Without this preservation radiation damage and heating effects can destroy the sample before complete data is recorded. The knowledge gained from this study will help the human and economic development of space, leading to a better understanding of fundamental physical, chemical and biological processes. Optimally preserved crystals, and the corresponding data from them, will produce more detailed structural knowledge of the processes of life benefiting molecular biology, medical and agricultural studies and help the development of new pharmaceuticals.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Two papers have been published since the last task progress report: 1. Snell, E.H. and Helliwell, J.R. Macromolecular Crystallization in Microgravity. Reports on Progress in Physics, 68, 799-853 (2005). 2. Finding a cold needle in a warm haystack: Infrared imaging applied to locating cryocooled crystals. Snell, E.H., van der Woerd, M.J., Miller, M.D. and Deacon, A.M. Journal of Applied Crystallography, 38, 69-77 (2005). The first is a detailed analysis of the physical basis of macromolecular crystallization in microgravity, the history, the successes, failures and future directions. The second covers the use of thermal imaging to locate crystals for high throughput structural genomics applications. Studies in progress include: 1. Cryocooling studied as a function of crystal volume. 2. Measurement of the heat generated by the interaction of a synchrotron X-ray beam with the crystal and the resultant thermal damage. 3. Development of a theoretical treatment explaining the heat deposition and predicting new ways to cool more efficiently. 4. Measurement of radiation damage in a sample and metrics that can be used to follow the damage process. 5. Testing of novel cryoprotectants and an investigation of potential free-radical scavenging. Papers are currently being prepared on several of these areas and will be presented as an invited talk at the Fourth International Workshop on X-ray Damage to Biological Crystalline Samples, which will be held at SPring8, Harima, Japan on March 7th and 8th, 2006.
Bibliography Type:	Description: (Last Updated: 12/14/2021)
Articles in Peer-reviewed Journals	Snell EH, Helliwell JR. "Macromolecular crystallization in microgravity." Reports on Progress in Physics 2005 Apr; 68(4):799-853. , Apr-2005
Articles in Peer-reviewed Journals	Snell EH, van der Woerd MJ, Miller MD, Deacon AM. "Finding a cold needle in a warm haystack: Infrared imaging applied to locating cryocooled crystals in loops." Journal of Applied Crystallography, 2005 Feb;38(1):69-77. , Feb-2005